

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4 HT	(a)	(i)		If a fault develops to make a <u>metal casing live</u> / <u>live</u> wire touches the <u>metal case</u> (1) it takes the current safely to ground / it prevents electric shock or protects the user / causes the fuse to blow or trips the mcb or rccb (1) Don't accept takes the electricity to ground or reference to electrified	2			2		
		(ii)		Faster acting or more sensitive and can be reset or reused - both required for the mark. Don't accept less expensive	1			1		
		(iii)		Total current = 24 A (1) which is split between the two routes [so no wire carries more than its rated value of 21 A] / which is less than 42 A / 12 A would travel each way (1)	2			2		
	(b)			Conduction in the saucepan base: Particles near the bottom vibrate <u>more</u> and pass their energy along (1) [Delocalised] electrons take energy through the metal pan (1) Convection in the water: Particles in the water move further apart (1) reducing density / so hot water rises / sets up a convection current (1)	4			4		
				Question 4 total	9	0	0	9	0	0